

World Motion Models: Flexible Sequence Modeling of SE(3) Trajectories

Jiahui Lei Qianqian Wang Trevor Darrell Angjoo Kanazawa
 UC Berkeley Harvard University UC Berkeley UC Berkeley
 {lei, trevordarrell, kanazawa}@berkeley.edu qwang@seas.harvard.edu

Abstract

Equipping artificial agents with spatial intelligence requires a comprehensive generative prior over the dynamic 3D world. We propose World Motion Models (WMMs) that capture “**what was, is, and will be where across time**” via sparse $SE(3)$ pose trajectories. WMMs are built on the observation that elements of dynamic scenes can be well approximated by a set of rigid $SE(3)$ trajectories, a minimal yet expressive primitive for 4D modeling. This representation unifies articulated objects, human bodies, hand-object interactions, piecewise-rigid scene dynamics, camera motion, and even robot states and actions into a single shared space. Given this representation, we cast the joint distribution of these entities as a flexible sequence modeling problem, utilizing flow-matching with per-token noise levels. Coupled with a context token mechanism for non-sequential conditioning, this formulation supports any-to-any marginal conditioning across an arbitrary number of entities and time steps. Tasks such as future prediction, motion infilling, model-predictive control, inverse kinematics, cross-embodiment retargeting, and policy learning all reduce to the application of different masks over the same network. Experiments on 6 diverse applications of 3D vision and robotics demonstrate the versatility and flexibility of WMMs with strong performance.

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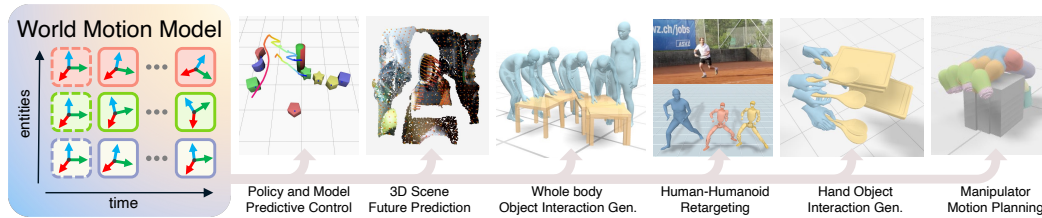


Figure 1: **World Motion Models** represent the 4D world across diverse entities and prediction modes in a unified manner, and can be flexibly conditioned for various downstream tasks.

1 Introduction

The world that surrounds us is inherently three-dimensional and highly dynamic—a 4D space where perception and interaction are fundamentally tied to movement. Physical actions are achieved by moving limbs or end-effectors, generating forces that translate into the motion of objects, while other agents dynamically navigate and act within this same environment. As humans, we naturally develop a strong prior over these moving entities: we intuitively anticipate how others will move, plan the physical motions required to achieve a desired goal, and foresee the physical consequences of our own actions. To act and reason in the physical world, intelligent systems benefit from the same

capability: a generative prior over the 4D world that captures the evolution of “what” and “where” across time.

In this work, we present a unified generative model which provides a general prior over the dynamic 3D world. Our approach is highly flexible with minimal assumptions and handles unstructured scenes and arbitrary object categories. By supporting various prompting modes, a single model enables diverse inference tasks ranging from world modeling (e.g., action-conditioned/free future prediction) to complex planning (e.g., motion infilling, inverse dynamics).

To build such a prior, we first need a unified representation that expresses the dynamic world in a physically grounded manner. Our insight is that complex, dense motion of the physical world can be compactly approximated by a sparse set of moving coordinate frames — specifically $SE(3)$ trajectories. Most entities of interest naturally reduce to this format: the dense flow of thousands of mesh vertices on a human or robot collapses into tens of joint trajectories [36, 60]; cameras and rigid objects track as single moving frames; and even non-rigid scene dynamics can be represented as interpolations of a small $SE(3)$ motion basis [43, 49, 85]. We therefore adopt the $SE(3)$ trajectory as our basic language for *what is where across time*. These $SE(3)$ trajectories subsume prior point trajectories in $\mathbb{R}^3$ but can be far sparser, replacing, for instance, the flow of thousands of mesh vertices with the tens of joint poses of a SMPL human [60] or a robot [36]. We further impose no category-specific structure: every entity is treated equally as a pose proxy, generalizing class-bound representations (SMPL [60], MANO [80], URDFs) to any unstructured scene. It is also orthogonal to video-based world modeling [3, 9, 70, 92, 103], which captures dynamics implicitly through color changes on a screen, entangling the underlying 3D motion with appearance, lighting, and viewpoint changes. In short, the $SE(3)$ trajectory provides a highly compact, physically grounded 3D abstraction for modeling arbitrary motion.

The question now becomes: how do we learn a generative prior over these $SE(3)$ trajectories? Since motion naturally unfolds over time, this is inherently a sequence modeling problem. Traditionally, this is treated as a strictly causal task: a standard world model or dynamics model predicts future states conditioned on past history and actions. However, humans can perform non-causal inference as well, such as determining the actions required to achieve a specific goal (inverse dynamics) or planning intermediate motions between waypoints. The inference type varies widely across applications. To support these diverse inference schemes, we propose World Motion Models (WMMs), flow-matching generative models with per-token noise levels. Inspired by Diffusion Forcing [15] and discrete diffusion [2, 69], WMMs treat each 6D $SE(3)$ pose at a given time step as an independent token, assign each its own noise level, and denoise the entire token field in a single shared forward pass.

Once trained, a WMM admits inference over arbitrary numbers of entities and time steps simply by dictating which tokens are clean (the condition) and which are noised (the prediction). Given past tokens, it acts as a future predictor $p(s'|s)$; applied to action tokens, it becomes a feed-forward policy $p(a|s)$; and when both past states and actions are provided, it acts as an action-conditioned dynamics model $p(s'|s, a)$ supporting model-predictive control. Ultimately, WMMs realize any $p(\text{unknown}|\text{known})$. By simply changing what is known, the exact same network transitions seamlessly between serving as a policy, an action prior, and a world model.

While standard trajectory modeling assumes a strictly consecutive temporal horizon, many practical tasks require reasoning beyond it. For example, planning a complex interaction often relies on an abstract goal state situated arbitrarily far in the future. Similarly, tasks like cross-embodiment retargeting require inferring correspondences between trajectories, such as those of a human and a robot. Providing a few paired demonstration frames from outside the current episode serves as an in-context template that tells the model “what is what.” To support these non-consecutive anchors, we enhance the flexibility of WMMs by introducing a set of *context $SE(3)$ tokens*. Rather than being bound to the continuous physical timeline of the current episode, they are assigned to a decoupled context timeline with its own learnable positional embeddings. Mechanically, these context tokens are treated identically to real trajectory tokens, meaning they can seamlessly condition the model or be generated themselves.

Together, this general $SE(3)$ formulation and our flexible sequence modeling enable WMMs to demonstrate broad versatility across applications, including robot policies and model-predictive control (MPC), scene future prediction, human-object interaction generation, and multi-embodiment humanoid retargeting (Sec. 3.1), all with SOTA performance.

Contributions. We propose a general formulation that uses the sparse $SE(3)$ trajectory as a unified representation for 4D world modeling. To learn a flexible generative prior over this space, we introduce WMMs, flexible flow-matching models with per-token noise levels. By supporting any-to-any marginal conditioning alongside a context token mechanism, WMMs deliver strong performance across diverse 3D vision and robotics tasks.

2 World Motion Models

2.1 Representation: A Language for Motion Modeling

World Motion Models aim to answer **what was, is, and will be where** across time. The thread connecting an entity at t_1 with the same entity at t_2 is its motion, and motion in real-world scenes is largely low-rank: a complex scene is driven by a relatively small set of *eigen* rigid trajectories, while dense spatial deformation arises as a local combination of those bases [43, 49, 85]. We assume the dynamic 3D world can be approximated by a finite set of rigid $SE(3)$ trajectories, with dense spatial deformation recovered by interpolation, skinning, or binary assignment. $SE(3)$ pose trajectories serve as the language with which we model the dynamic world. We leave modeling general linear local motion (scaling, shearing) to future work.

Coverage. As Fig. 2-(A) shows, this primitive approximates the motion of the principal subjects of 3D vision and robotics. *Scene future prediction* over 3D point trajectories [36, 46, 48] is the rotation-free special case of $SE(3)$ trajectories. *Human and hand-object motion* fits naturally: parametric body and hand models [47, 60, 80, 106] are sets of bone-joint $SE(3)$ frames, and articulated parts contribute additional rigid trajectories with per-part local shape. *Robot kinematics* treats each link as one rigid trajectory plus its mesh, the substrate for motion planning, retargeting, and control. *Robot state and action* are equally expressible whenever the controller (PD gains, etc.) is fixed, since the action then reduces to a set of virtual target poses. Thus, the same language captures both the observation and action of a standard control problem. *Camera motion* is, by definition, an $SE(3)$ trajectory of the viewpoint.

A pose at a time as a token. The elementary unit of our representation is a continuous token of “where”, given by a pose $T^{(l)} \in SE(3)$ and paired with a feature $F^{(l)} \in \mathbb{R}^D$ that supplies the “what” at time l . From now on we use $\{i, j, k\}$ to index entities and $\{l, m, n\}$ to index successive time steps. The feature is constant across time (shared along the trajectory) and is composed as $F_i = [f_i^{\text{sem}}; f_i^{\text{geo}}]$: the semantic part f_i^{sem} is either a DINO pixel feature [71] for video-prompted entities or a learnable embedding for coarse semantic categories (e.g. a body joint vs. an object). The

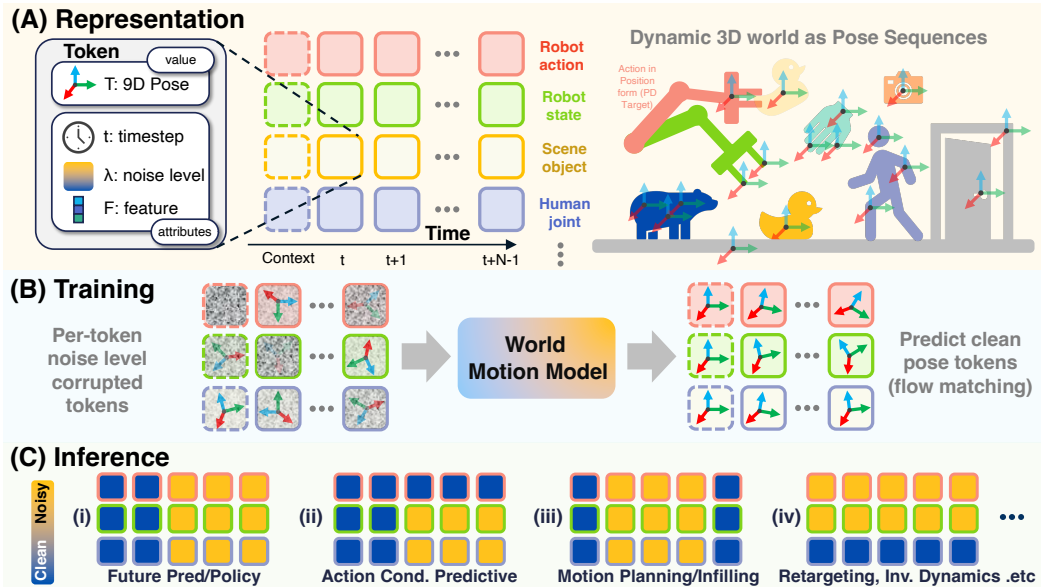


Figure 2: **World Motion Models:** (A) Dynamic scenes can be approximated by pose sequences. (B) During training, tokens corrupted at per-token noise levels are denoised to clean tokens. (C) During inference, various conditioning masks can be applied to one trained model.

geometric part f_i^{geo} is a small descriptor extracted by a PointNet [76] over the local point cloud when available (e.g. the local cloud of the object in human-object interaction or the shape of a robot body link) and is omitted otherwise (e.g. for a point prompt from a video). More details and ablations are in Appendix A.2. The pose token $T_i^{(l)}$ itself is parameterized as a 9-vector built from three points in $\mathbb{R}^3$,

$$T_i^{(l)} \equiv (\mathbf{x}, \mathbf{x} + \delta \mathbf{r}_1, \mathbf{x} + \delta \mathbf{r}_2) \in \mathbb{R}^9, \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^3$ is the position, $(\mathbf{r}_1, \mathbf{r}_2)$ are the first two columns of the rotation matrix under the 6D parameterization of Zhou et al. [104], and δ is a small fixed scalar (e.g. 0.2) empirically chosen to normalize the primal diffusion signal to a standard normal distribution (Appendix A.1). This 9-D form folds the unit-vector pair $(\mathbf{r}_1, \mathbf{r}_2)$ into the same $\mathbb{R}^3$ space as the translation, making normalization and training easier. Ground-truth labels for these poses at different time steps are widely available: they are read directly from parametric human reconstructions [27, 73] or from robot action recordings. For real videos, they can be computed using a closed-form approximation by defining a proper neighborhood around point traces produced by recent 4D reconstruction tools [36, 46, 48].

Dynamic scenes as trajectory sequence sets. An entity i moving through time produces a sequence of such pose tokens decorated with the same feature,

$$\tau_i = ([T_i^{(l)}, T_i^{(m)}, T_i^{(n)}, \dots]; F_i), \quad (2)$$

and a dynamic 3D scene is the set of M such trajectories,

$$\mathcal{X} = \{\tau_i, \tau_j, \tau_k, \dots\} = \{([T_i^{(l)}, T_i^{(m)}, T_i^{(n)}, \dots]; F_i)\}_{i=1}^M, \quad (3)$$

which we abbreviate as $\mathcal{X} = (\mathcal{T}, \mathcal{F})$ with the union of pose-sequences $\mathcal{T} = \{[T_i^{(l)}, T_i^{(m)}, T_i^{(n)}, \dots]\}_{i=1}^M$ and the corresponding feature set $\mathcal{F} = \{F_i, F_j, F_k, \dots\}$. Any dynamic scene over a causal temporal window can be expressed as such a pose-sequence set. However, many problems also benefit from non-causal context, e.g. a goal state far beyond the rollout window for an MPC controller to optimize toward, or a handful of motion-pair exemplars that specify a cross-embodiment retargeting relationship between sequences. We therefore extend each trajectory with a prefix of virtual time steps disjoint from the causal window,

$$\tau_i = ([\tilde{T}_i^{(a)}, \tilde{T}_i^{(b)}, \dots | T_i^{(l)}, T_i^{(m)}, T_i^{(n)}, \dots]; F_i), \quad (4)$$

where $a, b, \dots$ are virtual indices distinguished by a separate positional embedding (Sec. 2.4). Context tokens participate in training and inference exactly as causal ones, so a goal, an exemplar, or a label condition is simply another set of tokens. Without loss of generality, τ refers to this augmented form throughout, and Sec. 3 instantiates several choices of what to place in the virtual context.

Modeling target. A vanilla generative prior over this language captures $P(\mathcal{T} | \mathcal{F}, \mathcal{A})$, where $\mathcal{A}$ holds auxiliary conditions such as language or image tokens and $\mathcal{F}$ holds the per-token “what” decoration. Real applications, however, condition on arbitrary partial state rather than on $\mathcal{F}$ and $\mathcal{A}$ alone, so we instead study the family of any-marginal conditionals

$$P(\mathcal{T}_{\text{unk}} | \mathcal{T}_{\text{obs}}, \mathcal{F}, \mathcal{A}), \quad \forall \mathcal{T}_{\text{obs}} \subset \mathcal{T}, \quad \mathcal{T}_{\text{unk}} = \mathcal{T} \setminus \mathcal{T}_{\text{obs}}, \quad (5)$$

i.e., every distribution induced by partitioning the trajectory set into an observed subset and its complement. We treat this any-to-any property as the operational definition of a world model, and a single WMM supports it.

2.2 Training: Flow Matching with Per-Token Noise Level

We learn the any-marginal prior of Sec. 2.1 with a single flow-matching model under *asynchronous* noise levels (Fig. 2-(B)): each token (one entity at one time step) is assigned its own scalar level, sampled i.i.d. This is inspired by Diffusion Forcing [15] and diffusion large language models (dLLMs) [69, 93], which can model sequences in arbitrary order and over arbitrary subsets.

Forward process. Given a clean trajectory set $\mathcal{T}$ with M entities and N time steps (with pose tensor shape $M \times N \times 9$), we sample a per-token noise-level field $\Lambda \in [0, 1]^{M \times N}$ with each entry $\lambda_{i,l} \sim \mathcal{U}[0, 1]$ i.i.d., together with a Gaussian noise field $\epsilon \in \mathbb{R}^{M \times N \times 9}$, $\epsilon_{i,l} \sim \mathcal{N}(0, I_9)$. Each token moves along the path toward its sampled noise, with $\lambda_{i,l}$ broadcast across all 9 channels, giving the noised pose

$$T_i^{(l)} \Big|_{\lambda_{i,l}} = (1 - \lambda_{i,l}) T_i^{(l)} + \lambda_{i,l} \epsilon_{i,l}. \quad (6)$$

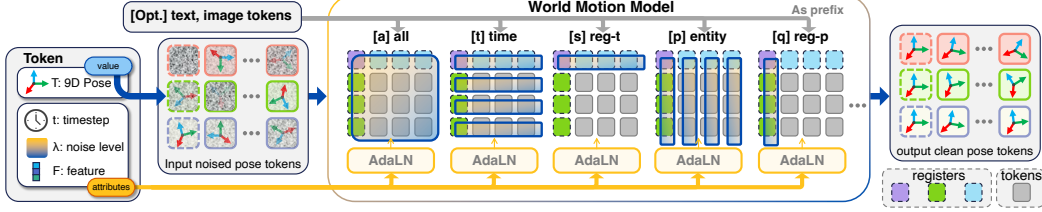


Figure 3: **Network:** The WMM architecture contains five factored attention types over a token grid padded with three register blocks, enabling efficient modeling (Sec. 2.4).

Velocity target and loss. Our model $\mathcal{M}_\theta$ is a velocity field in the flow-matching sense [54, 57] that predicts the velocity from the clean pose toward its sampled noise. The interpolation target is $v_{i,l}^* = \epsilon_{i,l} - T_i^{(l)}$, and, conditioned on the noised trajectory from Eq. 6, the feature set $\mathcal{F}$, the auxiliary condition $\mathcal{A}$, and the noise-level field Λ , the network is trained with a per-token error,

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathcal{T}, \Lambda, \epsilon} \sum_{i,l} \left\| \mathcal{M}_\theta(\mathcal{T}_\Lambda, \mathcal{F}, \mathcal{A}, \Lambda)_{i,l} - v_{i,l}^* \right\|_2^2. \quad (7)$$

Why this single objective is enough. Eq. (7) bakes the any-marginal prior of Sec. 2.1 into one network [15, 93]: each random Λ splits tokens into nearly-clean (observation-like) and nearly-noised (target-like), covering every conditional reverse process, with the deterministic observed/target split as the limiting case. Fixing Λ at inference (Sec. 2.3) samples the chosen marginal. The noise level scheduling can also be biased toward deployment patterns, e.g. more often scheduling a small λ on history and a large one on the future to sharpen forecasting while preserving coverage during training.

2.3 Inference: Various Marginal Distributions from the Same WMMs

A trained WMM is an entire family of conditional samplers under one set of weights. At inference, the user picks a per-token mask of observed and target tokens, sets $\lambda_{i,l} = 0$ on the former and $\lambda_{i,l} = 1$ on the latter, and integrates $\mathcal{M}_\theta$ from $\lambda = 1$ down to 0: observed tokens stay as input, while target tokens are transported by the velocity field, yielding a sample from $P(\mathcal{T}_{\text{unk}} \mid \mathcal{T}_{\text{obs}}, \mathcal{F}, \mathcal{A})$. Different masks cover various applications (Fig. 2-C), for example:

- (i) **Future prediction and policy.** $\lambda_{i,l} = 0$ for $l \leq l_0$, $\lambda_{i,l} = 1$ for $l > l_0$. State-only entities give future prediction; with action entities in $\mathcal{T}$, the same mask is a robot policy.
- (ii) **Action-conditioned predictive model (forward dynamics).** Action entities known, future states unknown: a learned forward simulator for model-predictive control.
- (iii) **Motion planning and in-filling.** Sparse keyframes (possibly partial) are observed; the rest are targets: robot motion planning and motion interpolation.
- (iv) **Retargeting, partner prediction, and inverse dynamics.** Partition by entity: $\lambda_{i,l} = 0$ for $i \in \mathcal{O}$, $\lambda_{i,l} = 1$ otherwise. Taking $\mathcal{O}$ to be one body gives cross-embodiment retargeting; taking it to be a hand gives hand-object HOI partner prediction (or vice versa); taking it to be the state entities gives inverse dynamics.

Note that any known–unknown pattern is supported, beyond the examples mentioned above. Such a design has additional benefits: **robustness to observation noise.** Real observations are imperfect, and WMMs accommodate this by setting $\lambda_{i,l}$ on observed tokens to a small positive value (e.g. 0.05) rather than exactly 0, which turns each observation into a strong hint and lets the integrator slightly rectify the token along the velocity field as the noise level decreases. **Generalization to many sequences:** RoPE in a transformer already helps the model generalize beyond the number of sequences seen during training. A simple inference-time algorithm (App. B.2) extends this generalization further when combined with the per-token noise formulation, enabling dense trajectory generation well beyond the training count.

2.4 Implementation: Model Architecture

As shown in Fig. 3, our model supports arbitrary numbers of entities and time steps while addressing three issues: (i) *per-token noise levels:* $\lambda_{i,l}$ has to be specified *per token*, not as a single global scalar as in a conventional flow-matching backbone [54, 74]; (ii) *scale:* a modest 30-frame, 400-trajectory scene already contains 12K tokens, on which dense self-attention is expensive; (iii) *missing*

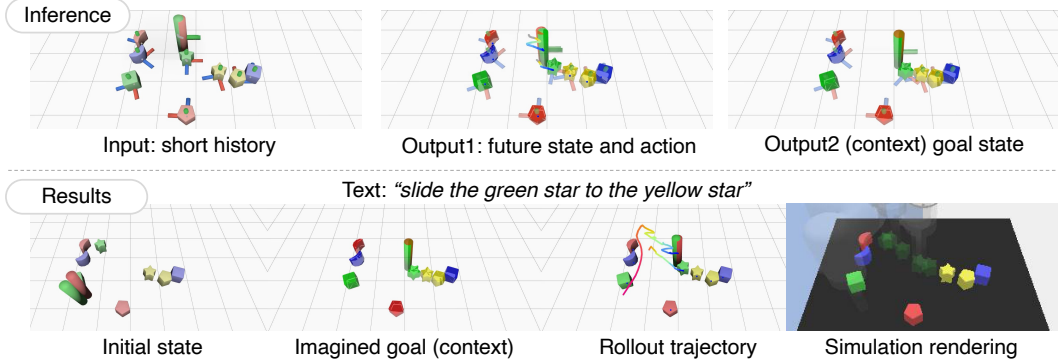


Figure 4: **Language Table.** Top: inference pattern; history is the input, while future and goal are predicted by WMMs. Bottom: policy rollout results.

Table 1: **Language Table.** Per-task and macro-average success rate (%).

Method	B2B↑	B2AL↑	B2BRL↑	B2RL↑	Sep↑	Avg↑
Scratch [94]	4.0	6.0	10.0	6.0	52.0	15.6
UniPi [24]	14.0	4.0	12.0	10.0	72.0	22.4
VPT [6]	36.0	38.0	48.0	26.0	70.0	43.6
LAPA [94]	58.0	56.0	52.0	48.0	96.0	62.0
ActionVLA [94]	76.0	72.0	76.0	70.0	90.0	76.8
LCB [81]	90.0	71.0	66.0	73.0	99.0	80.0
LangTable orig. [62]	88.0	65.0	70.0	68.0	94.0	77.0
WMM	90.0	76.0	80.0	90.0	100.0	87.2

Table 2: **Plain Policy versus MPC.** On Language Table absolute tasks.

	B2B		B2AL	
	Plain	MPC	Plain	MPC
Succ.↑	90.0	100.0	76.0	80.0
Mean steps↓	79.5	79.1	92.7	87.2
Med. steps↓	73.0	68.5	87.0	85.5
Min steps↓	26	16	14	2
Max steps↓	198	175	187	171

observations: real data come with invalid intervals (occlusion, truncation, tracker drop-out) that cannot simply be masked during training. We address each as follows: **(1) Per-token conditioning via AdaLN.** For each token (i, l) we concatenate (a) a sinusoidal embedding of $\lambda_{i,l}$, (b) the feature F , and (c) a pooled embedding of the auxiliary tokens $\mathcal{A}$ (e.g. a language summary). A per-block wide multi-layer FFN maps this vector to per-token AdaLN scale-and-shift parameters that modulate every transformer block [74]. **(2) Registers and factored attention.** We pad the $M \times N$ token grid with three blocks of learnable registers [21] and apply five attention types on the padded grid: a self-attends across all tokens; τ and p are time- and entity-axis self-attention; s and q are register-only counterparts on the same two axes, widened aggressively in channels at negligible runtime cost to add expressivity. **(3) Training under missing observations.** Invalid time steps must still participate in attention, since at inference the missing mask is unknown. We pin invalid tokens to $\lambda_{i,l} = 1$ (pure noise) and sum the loss in Eq. (7) only over tokens with ground-truth labels. Virtual-context tokens (Sec. 2.1) are flagged by a 1-bit prefix mask and given learnable position embeddings. Text or image tokens attach as an attention prefix.

3 Experiments

We evaluate WMMs on open-source benchmarks spanning policy learning, future prediction, HOI, and retargeting, plus a set of ablations. We train a separate WMM checkpoint on each benchmark’s training split; joint training of one WMM across all benchmarks is left to future work. The entities modeled by a WMM are defined according to the task: examples include a link or joint of a robot, human body, or object (Secs. 3.1, 3.4, and 3.3), or any prompted query point in the general background (Sec. 3.2); see below for details.

3.1 Language Table: Policy Learning and Forward Dynamics

Language Table [62] is a tabletop pushing benchmark with five task families. We follow the evaluation protocol of LAPA [94] and LCB [81]: 50 rollouts per family on the eight-block setup with a 200-step horizon, reporting per-family and macro-average success rates. **Setup.** As in Fig. 4, the entity set comprises the 6D pose of every block together with the robot end-effector state (green) and action target (red); the context tokens are configured to capture the final frame of a successful trajectory. The feature descriptor of each token is defined as follows: F_{sem} consists of three learnable embeddings

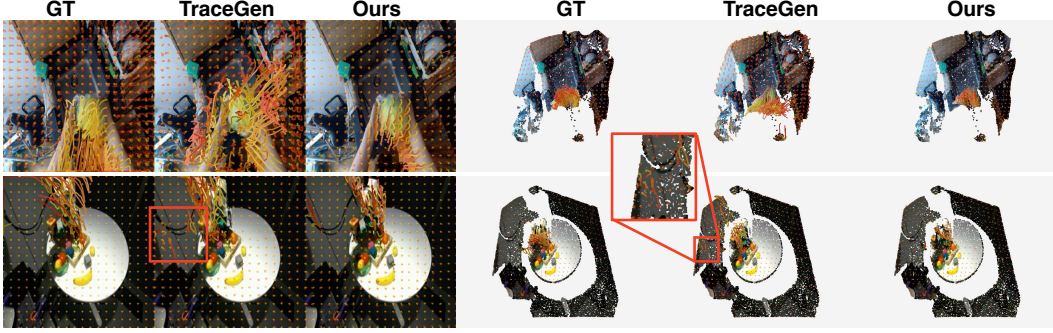


Figure 5: **TraceGen**. Qualitative results on the TraceGen benchmark [46].

Table 3: **TraceGen**. 3D-trajectory metrics on the Droid and EpicKitchen test splits.

Method	Dr MSE↓	Dr MAE↓	Dr Endp.↓	Ep MSE↓	Ep MAE↓	Ep Endp.↓
TraceGen [46]	0.206	1.289	0.285	0.445	2.721	0.791
WMM, per-dataset ($P=128$)	0.151	1.129	0.200	0.322	2.130	0.556
WMM, per-dataset ($P=400$)	0.178	1.169	0.236	0.329	2.167	0.580
WMM-Joint ($P=128$)	0.167	1.172	0.225	0.346	2.256	0.615
WMM-Joint ($P=400$)	0.187	1.179	0.259	0.351	2.287	0.621

for object, link-state, and link-action, while F_{geo} is obtained by applying a PointNet to the local point cloud of the object or link. We train one model on the public oracle trajectories across all five tasks, and inference runs in the policy mode of Sec. 2.3: given a five-step history and language prompt, a ten-step action chunk is read out from the action entity in the future prediction, ensembled across queries as in ACT [100]. The context (goal state) is also generated from the language and observation. **Results.** As in Tab. 1 and Fig. 4, a WMM obtains the strongest macro-average, demonstrating that WMMs are effective and accurate as policies.

Action optimization with MPC. WMMs can also be used as a differentiable world model rather than a plain policy. For tasks with absolute-position instructions (B2B, B2AL) in LangTable, given the generated goal state (via context tokens), we can apply model predictive control (MPC) to optimize the predicted action so as to maximize a heuristically defined value function: the similarity between the final frame of the action chunk and the generated goal. Since our model is fully differentiable, gradient-based MPC can be hybridized with sampling-based methods (full procedure in App. B.3); Tab. 2 compares the plain policy with the MPC-optimized policy on B2B and B2AL, indicating both a higher success rate and faster rollout to success.

3.2 TraceGen: Dense 3D Scene Future Prediction

TraceGen [46] is a 3D trajectory generation benchmark in which the model takes a single RGB-D start frame and a language instruction and predicts the motion of 400 keypoints over 32 steps. We follow its protocol on the two subsets that provide 3D trajectories, Droid and EpicKitchen, reporting the trajectory-generation errors MSE, MAE, and endpoint MSE (each $\times 100$). **Setup.** Each of the 400 keypoints is one entity carrying a 3D position and an orientation (computed by local Procrustes in the dataset), starting at the first frame ($t=0$). We train four WMM checkpoints over the experimental space (Droid only, EpicKitchen only, and joint on both) at two keypoint window sizes ($P=128$ and $P=400$, where P is the number of trajectories seen during training). No special context token is used in this experiment. **Results.** As shown in Tab. 3 and Fig. 5, WMM achieves the best performance in every column in both environments. Note that the $P=128$ variant is trained with 128 randomly selected trajectories, meaning that it can accept random queries during inference. Although only 128 trajectories are observed in parallel during training, the model can directly infer all 400 trajectories and even achieves better performance ($P=128$ wins all metrics); we hypothesize that this stems from the random sampling of 128 out of 400 trajectories during training, which prevents overfitting. The experiments also show that joint training across both domains is feasible, suggesting that future work can pursue a single unified model across domains.

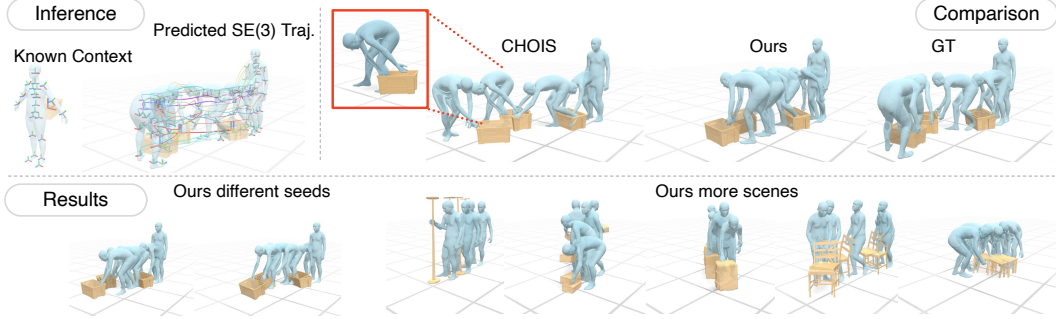


Figure 6: **OMOMO**. Qualitative results on OMOMO [50]. Top left: inference pattern. Top right: comparison with CHOIS [51]. Bottom: additional results from WMMs.

Table 4: **OMOMO**. Results under the CHOIS [51] protocol.

Method	$T_s \downarrow$	$T_e \downarrow$	$T_{xy} \downarrow$	$H_{feet} \downarrow$	$FS \downarrow$	$C_{prec} \uparrow$	$C_{rec} \uparrow$	$C_{F1} \uparrow$	$C_{\%} \uparrow$	$MPJPE \downarrow$	$T_{root} \downarrow$	$T_{obj} \downarrow$	$O_{obj} \downarrow$
InterDiff [90]	–	158.84	72.72	0.90	0.42	0.63	0.28	0.33	0.27	25.91	63.44	88.35	1.65
MDM [83]	5.18	33.07	19.42	6.72	0.48	0.72	0.47	0.53	0.43	17.86	34.16	24.46	1.85
Pred-OMOMO [50]	2.39	8.03	4.15	7.08	0.40	0.73	0.66	0.66	0.62	18.66	28.39	16.36	1.05
CHOIS w/o L_{obj} [51]	5.76	14.16	8.44	6.55	0.40	0.75	0.50	0.55	0.43	14.34	21.97	15.53	0.98
CHOIS w/o F_{all} [51]	1.75	6.61	2.69	6.64	0.38	0.78	0.49	0.55	0.41	15.23	24.13	11.51	0.99
CHOIS [51]	1.71	6.31	2.87	4.20	0.35	0.80	0.64	0.67	0.54	15.30	24.43	12.53	0.99
WMM	0.15	6.48	4.40	0.84	0.56	0.79	0.67	0.76	0.59	12.07	19.87	16.53	1.03

3.3 OMOMO: Human-Object Interaction

OMOMO [50] is a benchmark for object-conditioned full-body human-object interaction synthesis: given a partial object keyframe and a language prompt, the model generates the body and object motion. We follow the protocol of CHOIS [51]. **Setup.** The entity set comprises the SMPL body joints and the interaction object’s rigid poses. The Context tokens are always set to a SMPL human in an A-pose (which conveys entity identities) and the object at the origin. The time window is set to the dataset’s full length of 120 frames.

Note that in this human body example and the following humanoid and hand experiments, WMM directly models pose trajectories even when a structure such as the SMPL model is known, enabling WMM to model general and diverse morphologies within a single architecture and model. The following visualizations are produced by directly animating the meshes with linear blend skinning (LBS) or rigid transformations using the poses predicted by WMM, showing the accuracy of our model. To conduct a fair comparison (e.g., baselines use SMPL parameters to compute metrics), we convert the output of WMM to the corresponding parametric space; further discussion of this projection is provided in Appendix A.3.

Results. Tab. 4 and Fig. 6 show comparable or even better performance on the HOI synthesis task. Fig. 6 further shows that when CHOIS produces implausible predictions in which the human intersects the object, the WMM maintains plausibility and can generate diverse motions. Additionally, to compare model size and efficiency, we measure the parameter counts of TraceGen and WMM (674.6M vs 438.8M) and their inference times for 400 tracks over 32 steps (2.66 s vs 4.18 s).

3.4 Human and Humanoid: Kinematic Retargeting

We construct a benchmark that takes a SMPL human motion sequence and produces a Unitree G1 humanoid [84] trajectory. We evaluate retargeting quality by feeding the trajectory to a low-level robot controller (TWIST [97]), rolling it out in physics simulation, and measuring tracking error and success rate. Three input settings probe WMMs’ flexibility: clean human input at 30 FPS; sparse keyframes at 3 FPS that the model must infill; and human input with noise at every frame ($\sigma=0.1$ on axis-angle and root translation). **Setup.** The entity set comprises the SMPL human and the G1 humanoid links, paired across embodiments inside the Context-token exemplar frames (Sec. 2.1; two samples in Fig. 7). A WMM is trained on a mix of AMASS [64] and Embod3D [66]; a single checkpoint serves all three settings. The humanoid’s kinematic reference (ground truth) is obtained by running and filtering PHUMA [45], a slow offline optimization method. **Results.** Tab. 5 and Fig. 7

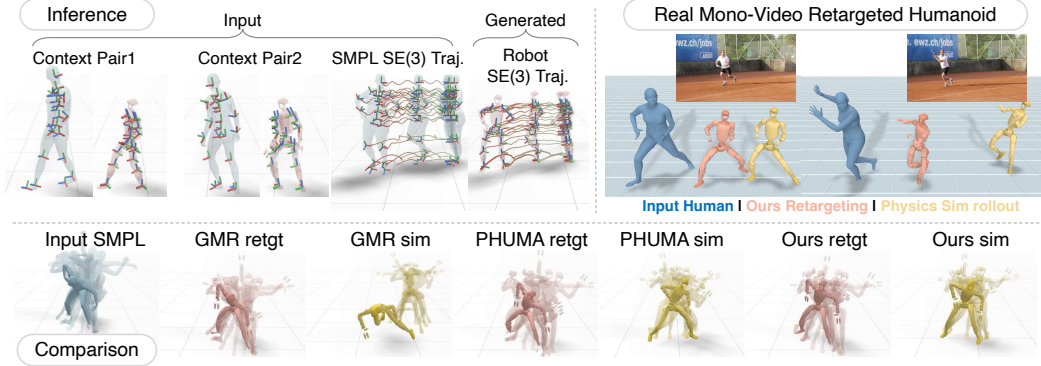


Figure 7: **Human-to-Humanoid Retargeting**. Top left: inference pattern. Top right: retargeting from real-world video. Bottom: comparison against baselines.

Table 5: **Humanoid Retargeting**. Results under three conditioning regimes (clean, sparse-keyframe in-fill, per-frame noised).

Method	Clean (30 FPS)				In-fill (3 FPS keyframes)			Noisy ($\sigma=0.1$)		
	FPS $\uparrow$	SR-seq $\uparrow$	gerr $\downarrow$	lerr $\downarrow$	SR-seq $\uparrow$	gerr $\downarrow$	lerr $\downarrow$	SR-seq $\uparrow$	gerr $\downarrow$	lerr $\downarrow$
GMR [1]	35.03	0.7945	0.6911	0.0651	0.8095	0.6542	0.0592	0.7160	1.0196	0.1158
PHUMA [45]	2.27	0.8255	0.5509	0.0530	0.6535	1.3744	0.0745	0.8060	0.7269	0.0703
WMM	86.97	0.8690	0.5938	0.0420	0.8525	0.6371	0.0518	0.8530	0.7785	0.0691

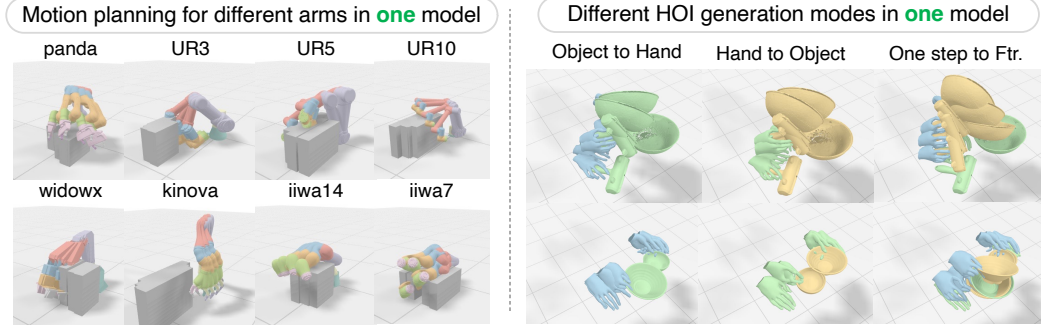


Figure 8: **More Applications**. Qualitative results of WMMs on additional tasks.

report the comparison against GMR [1] and PHUMA [45]. GMR is a fast, real-time method but provides limited quality; PHUMA is an offline optimization method that produces more physically plausible results (less overlap, etc.) but is slow. The resulting WMM is a retargeting prior that is fast (note the throughput reported in the first column) and learns a prior over physically plausible, safe motion across sequences, providing higher-quality and safer retargeting motion. Fig. 7 also shows real-world monocular-video-based humanoid retargeting and control results.

3.5 More Applications

The left side of Fig. 8 shows an additional application of WMMs: motion planning with known first and last timesteps (Fig. 2-(C ii), labelled by PyRoKi [42]). This works across more than 8 different manipulators within a single model, showing that our flexible modeling techniques can model diverse kinematic targets across morphologies. The right side of Fig. 8 shows hand-object interaction generation on TACO [58] under various known-unknown patterns within one model (the green mesh denotes the known condition), showing that WMM can be used in diverse ways during inference.

3.6 Ablations

We ablate per-block AdaLN and the per-token-noise schedule (Sec. 2.2) on Language Table B2B, and architectural primitives (registers, factored axis-wise attention; Sec. 2.4) on TraceGen EpicKitchen. Each ablation toggles one field; details in App. B.4. **Language Table**. Tab. 6 (left): removing AdaLN

drops B2B success by 6 pp but the model still learns a competent policy; removing per-token noise (using synced noise with guidance-based conditioning) yields no successful rollout, since policy mode requires very high accuracy. **TraceGen.** Tab. 6 (right): collapsing the middle three layers to plain DiT (no efficient register) degrades performance because the same hardware then supports a smaller batch size and slower training. Collapsing all seven to plain DiT (no axis-wise factorization) yields a $\sim 15\times$ MAE regression, as the model loses the explicit temporal-spatial inductive bias.

4 Related Work

Motion Generation and World Models.

Existing generative motion models are fragmented by actor class through morphology-specific parameterizations [41, 53, 83], while general unified approaches [36, 48, 70] operate in dense observation spaces such as pixel-aligned point tracks, 3D flow, or video entangling 3D dynamics with appearance, or solely in 2D pixel motion space [68, 77, 78]. A piecewise-rigid lineage uses $SE(3)$ transforms [10, 33] but is restricted to single-agent or fixed-object settings. WMMs instead encode every moving unit (robot link, body part, scene primitive) as a (sparse) set of $SE(3)$ tokens in a shared abstraction, leading to a potentially more unified and shared motion prior over all moving subjects in the physical world.

Flexible Policies and Sequence Modeling.

Recent work couples world modeling with policy via video-action models [52, 56, 105] or latent-dynamics planners [28, 29], but ties the world model to a fixed morphology or pixel tokens. Diffusion Forcing [15] and discrete masked diffusion [2] introduce per-token noise schedules for flexible conditioning. Specifically, Diffusion Forcing [15] is the modeling method closest to ours. Although theoretically general, it primarily focuses on temporal scheduling across sequence frames. Our method generalizes the idea of per-token noise to spatial and temporal axes, and our architectural design makes this practical for more than 10K tokens. We provide a unified representation of heterogeneous $SE(3)$ trajectories and add a decoupled context-token mechanism, so one network spans prediction, policy rollout, and non-causal planning. See App. B.1 for an extended discussion.

5 Conclusion

We propose World Motion Models, a unified generative prior over the dynamic 3D world via sparse $SE(3)$ pose trajectories. A per-token-noise-level flow-matching model with context tokens supports any-marginal conditional inference from one network, attaining state-of-the-art results on policy learning, scene future prediction, human-object interaction, and humanoid retargeting.

Limitations. (i) We train one WMM checkpoint per benchmark; joint training across domains and increasing the amount of supervision data are left to future work. (ii) Generalizing $SE(3)$ to more general linear groups is also a future direction. (iii) The current framework requires geometry (e.g., depth and estimated current and past states) as input; relaxing these requirements and enabling direct inference from RGB videos are also directions for future work. (iv) Another limitation of the current model is that it does not generalize to novel morphologies unseen during training; further investigation of in-context learning with our proposed context mechanism may suggest solutions in the future.

Broader impact. WMM may have benefits in robotics, mixed reality, animation, games, and simulation. We do not see major direct negative ethical or societal consequences of our work.

Table 6: **Ablations.** Language Table B2B (left), TraceGen EpicKitchen (right).

Variant	SR $\uparrow$	Variant	MAE $\downarrow$	MSE $\downarrow$	Endp. $\downarrow$
WMM	0.860	WMM	2.167	0.329	0.580
no AdaLN	0.800	no eff.	2.356	0.344	0.602
no per-tk	0.000	plain DiT	32.05	16.97	17.64

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A More Ablations

A.1 Rotation scale δ

We vary the rotation scale δ for humanoid retargeting following the setup of Tab. 5. The choice $\delta = 0.2$ reported in Sec. 2.1 of the main paper is empirical and is intended to keep the primal diffusion signal approximately normalized around mean 0 and standard deviation 1. Table 7 compares this choice with $\delta = 0.6$ and $\delta = 1.0$.

Table 7: Rotation-scale ablation on humanoid retargeting.

Setting	SR-seq $\uparrow$	gerr $\downarrow$	lerr $\downarrow$
Paper ($\delta = 0.2$)	0.8690	0.5938	0.0420
$\delta = 0.6$	0.8645	0.6440	0.0449
$\delta = 1.0$	0.8030	0.7709	0.0504

A.2 Semantic and geometric features

We ablate F_{sem} and F_{geo} by setting the corresponding features to zero.

Language Table. Here, F_{sem} consists of three learnable embeddings for object, link-state, and link-action, while F_{geo} is obtained from a PointNet on the local point cloud of the object or link. Removing F_{geo} substantially reduces the success rate (Table 8). The local colored point cloud grounds the language command in specific objects, whereas the semantic feature identifies the entity category rather than which object is which.

Table 8: Feature ablations on Language Table. noFsem and noFgeo zero the respective features; noBoth zeros both.

	Paper	noFsem	noFgeo	noBoth
Avg Succ Rate $\uparrow$	0.8720	0.8560	0.1960	0.1680

TraceGen. Here, F_{sem} is the DINO feature queried at the pixel location. No F_{geo} is used because the prompt is a point. Removing DINO degrades performance on both EPIC and DROID (Table 9), although the model can still predict future motion from the available geometric information. Semantic features encode “what” the entity is, improving the expressivity of the network.

Table 9: Semantic-feature ablation on TraceGen.

Setting	EPIC			DROID		
	MAE $\downarrow$	MSE $\downarrow$	Endpoint $\downarrow$	MAE $\downarrow$	MSE $\downarrow$	Endpoint $\downarrow$
Paper	2.2564	0.3458	0.6146	1.1718	0.1665	0.2253
noFsem	2.4032	0.3627	0.6148	1.2894	0.1876	0.2539

A.3 Consistency with parametric models

For evaluation, we convert predicted poses to each baseline’s format: SMPL joint rotations computed from the $SE(3)$ poses, or robot joint angles obtained from relative axis-angles along the kinematic chain with one degree of freedom per joint. Evaluation then follows the corresponding baseline protocol. We measure the consistency between the raw $SE(3)$ predictions and their projected fits in two settings.

SMPL on the OMOMO test set. We compare raw $SE(3)$ joints and bones with the forward-kinematics skeleton of the projected SMPL parameters.

Table 10: Self-consistency of SMPL predictions on the OMOMO test set.

Bone length error	Bone direction error	Raw vs. projected MPJPE
7.21 mm	2.99°	19.15 mm

G1 humanoid test set. We measure the off-hinge-axis rotation and translation removed by projection, together with the difference between raw and forward-kinematics link positions.

Table 11: Self-consistency of predictions on the G1 humanoid test set.

Off-hinge-axis rotation	Off-hinge translation	Raw vs. projected MPJPE
3.57°	13.37 mm	31.38 mm

A.4 Data-generation cost

For parametric and known embodiments, including SMPL bodies, MANO hands, and known robot URDFs, joint or link $SE(3)$ poses can be read directly from saved parameters or actions.

To obtain pose labels for supervision from general RGB videos, we first preprocess the videos to obtain 3D tracks. Table 12 outlines the procedure and reports measurements for a video with 40 frames, 32,768 tracks, and resolution 378×504 . The reported total includes the optional full-video object segmentation (VOS) stage, which has not yet been used in this paper.

After the tracking stage, we compute local frames from these tracks. We first define a KNN topology using a topology-aware curve distance: the maximum distance between two trajectories over the times when they are co-visible. Local rigid motion is then obtained by solving a closed-form Procrustes problem.

Table 12: RGB-video-to-3D-track processing cost.

Steps	Time, VRAM
Calibration: resize and run GeoCalib on key frames to undistort intrinsics and estimate the gravity/vertical direction.	4.3 s / 5,856 MB
Geometry: run chunked slice-and-align DA3 (to keep VRAM bounded on long sequences) to estimate depth and camera parameters; also run DA3-metric on key frames to compute the metric scale.	34.6 s / 14,137 MB
Tracking: run 2D tracking using TAPNext and multiple rounds of motion-aware resampling (to keep VRAM bounded and focus more on moving areas), then lift the 2D tracks to 3D at visible time steps using the camera parameters and depth, with depth-boundary and anti-jitter filtering.	64.6 s / 8,151 MB
[OPT] Full VOS (not used in the paper): our pipeline also runs DEVA in auto-mode to segment and track anything in the full video.	27.2 s / 14,079 MB
Total cost	131 s / 14.1 GB

B Technical appendices and supplementary material

B.1 Related Work

Motion Generation and 4D World Modeling. Generative motion modeling has historically fragmented along actor classes. Body, human-human interaction, hand-object interaction, and scene dynamics each use their own state parameterization and task interface [7, 23, 41, 51, 83, 90]. A more recent line of work on unification scales LM backbones, million-scale corpora, compositional priors, and motion-understanding foundations on human motion alone [4, 53, 67, 72, 87, 91, 99], and three further papers consolidate human-side tasks under a single prior. MAGNet [65] models multi-agent body behavior with diffusion-forced canonical pairwise transforms, UMO [20] casts object-aware

human motion as a unified in-context problem, and FUSION [25] trains an unconditional SMPL-X body and hand prior. All of these inherit a body-parametric state and so remain actor-homogeneous within the human family. A second line writes motion into a shared observation space that crosses morphologies. MoMap [48] diffuses pixel-aligned dense 3D motion maps; PointWorld [36] rolls point cloud and gripper flow as a forward simulator; TraceGen [46] flow-matches camera-frame 3D traces conditioned on a single RGB-D start frame and language; GeneralFlow [96] predicts dense 3D flow from RGB-D; EC-Flow [18] maps embodiment-centric pixel flow to end-effector SE(3) via inverse kinematics; ATM [88] pretrains 2D point tracks as a representation for behavior cloning; TAPNext [102] casts point tracking itself as causal next-token prediction; and sparse-track or sparse-observation diffusion also extends to dexterous manipulation and SMPL completion [13, 44]. These methods share an observation space across actors, but the underlying tokens are translation-only points or dense flow that carry no per-entity identity and discard the rigid-body rotation that drives contact and articulation. A third line generates ambient 4D scene dynamics with explicit trajectories: Comp4D [89] factorizes a 4D world into per-object trajectories plus per-object deformation, and TC4D [5] conditions 4D generation on explicit SE(3) trajectories, but both target per-scene synthesis rather than a generative prior over heterogeneous moving entities. A piecewise-rigid lineage models scenes directly as per-part SE(3) transforms: SE3-Nets [10] and SE3-Pose-Nets [11] warp tabletop point clouds through K learned rigid motions conditioned on one robot’s actions, OASIS [16] predicts the end-effector pose trajectory for manipulation, and SPOT [33] denoises source-target object pose trajectories from a few demonstrations, yet each remains a single-agent system over a fixed rigid-object set tied to one manipuland or one robot. The closest points of comparison to WMMs are therefore MAGNet and UMO (unified conditioning, human-only), TraceGen and MoMap (trajectory-space world modeling without rotation or per-entity identity), and SPOT together with SE3-Nets (SE(3)-trajectory denoising on a fixed manipuland set), each of which stays inside one of these regimes. WMMs instead encode every moving unit, whether a robot link, hand, body, object, or scene primitive, as a sparse SE(3) token in a shared rot6d+translation space (Sec. 2.1), so one denoiser is promptable across actor classes and queries arbitrary entities at arbitrary time steps, rather than tracking a fixed dense grid of points or a single morphology family.

World Models and Robot Policy. A unified motion substrate must also support rollout and policy. Pixel-space world-action models keep the substrate in 2D video: DreamDojo [26], DreamZero [95], and Cosmos [70] synthesize frames or extract actions from pretrained video diffusion backbones, and World-VLA-Loop [56] closes the action-perception loop on this pixel substrate with a vision-language-action policy. A related dense-observation model, DyWA [63], runs dynamics-adaptive world-action coupling over single-view point clouds for non-prehensile manipulation, swapping pixel tokens for a 3D point cloud but keeping the world model in a dense observation space. Latent-dynamics planners abstract observations into a learned latent representation: Hierarchical Humanoid WM [29] and DreamerV3 [28] couple latent rollouts with model-predictive control or value-based control, but each instance hard-codes the morphology or the target RL environment. Two recent approaches directly model heterogeneous forward dynamics: Cross-Embodiment WM [30] runs particle-graph rollouts and Geom4D-Robot [59] jointly denoises RGB and pointmap, yet both recover SE(3) only post-hoc through an additional pose-extraction step rather than treating SE(3) as the modeling primitive; imagination-policy goal generators [34] and long-horizon scene or traffic generators [75, 98] likewise stay morphology- or domain-locked. Vision-language-action policies form a complementary set of output interfaces over various tokenizations of action [8, 79, 101], and Moto [17] bridges video pretraining to action through a latent motion-token language. Most closely related to our unification approach, UWM [105] and UVA [52] jointly denoise video and action and pursue multi-axis flexibility through shared backbones, but both still tie the world-model side to pixel tokens. WMMs close this gap by predicting SE(3) trajectories of every tracked entity under a per-token conditioning mask, so one flow-matching model with per-token noise serves as a policy, predictor, and world simulator across arm, humanoid, dexterous hand, human, and tabletop settings, with no morphology-specific head and no separate world and policy modules (Sec. 2.3).

Sequence Modeling with Flexible Conditioning. Diffusion forcing [15] assigns each token an independent noise level, so any subset of tokens can be revealed, in-filled, or rolled out from one trained network, and CausalFusion [22] reconciles autoregressive and diffusion training under the same per-token schedule. This primitive has propagated along several tracks: streaming video adopts diffusion forcing for long-horizon rollouts and live generation [35, 37, 55]; motion and robot control adopt the same schedule on body, action, or interaction tokens [12, 32, 38, 39]; discrete

and masked sequence models reach the same flexible-conditioning behavior through token-level masking schedules [2, 14, 86]; a separate trajectory-level denoising line conditions whole state-action plans on goals or rewards [19, 40]; and earlier flexible-conditioning recipes provide classifier-free guidance, masked image inpainting, and masked motion completion [31, 61, 82]. Each of these instances, however, binds the per-token schedule to a single domain: video frames, one actor’s skeletal tokens, one morphology’s action horizon, or text. WMMs apply the same per-token noise schedule to SE(3) trajectories of heterogeneous entities (Sec. 2.2) and extend the trajectory with a context token prefix on a separate timeline for non-causal anchors such as goal states or retargeting exemplars (Sec. 2.1), so one trained network selects between motion synthesis, future prediction, policy rollout, and planning based solely on which tokens are noised at inference (Sec. 2.3).

B.2 Dense generation beyond the trained budget

WMMs are trained on a bounded number of entities and time steps, yet many uses require more entities or time steps than fit in one forward pass (even though RoPE helps to generalize beyond the context window): long-horizon rollouts, dense scene flow, and multi-agent simulations. Splitting into independent chunks fails because two chunks denoised independently disagree at their interface. The *anchored sliding-window* sampler in Alg. 1 instead exploits asynchronous-noise training to keep partially denoised tokens of one chunk as fixed anchors when denoising the next. With a context size S and an anchor count $A < S$, the sampler descends a discrete schedule $\lambda_K > \dots > \lambda_0 = 0$, so that every forward pass sees S entities at consistent asynchronous levels, exactly the regime used in training with (7). The A shared anchors stitch successive batches into a globally consistent rollout. Asynchronous training plus a non-grid token interface together unlock arbitrarily many entities and time steps at inference.

Algorithm 1 Anchored sliding-window dense sampler.

Require: $S, A < S, \lambda_K > \dots > \lambda_0 = 0, \mathcal{E}$ at λ_K

- 1: **for** $k = K, \dots, 1$ **do**
- 2: $\mathcal{D} \leftarrow \emptyset$
- 3: pick S at λ_k ; advance to λ_{k-1} ; add to $\mathcal{D}$
- 4: **while** any entity at λ_k **do**
- 5: A anchors $\sim \mathcal{D}$; $S-A$ fresh at λ_k
- 6: advance fresh to λ_{k-1} ; add to $\mathcal{D}$
- 7: **end while**
- 8: **end for**
- 9: **return** $\mathcal{E}$ at $\lambda_0 = 0$

B.3 MPC details

At each ACT replanning step, the MPC mode will (i) run one text-conditioned plain inference pass to obtain an imagined goal trace; (ii) build $K=4$ candidate action chunks of length 10 (one baseline plus three perturbed by Gaussian noise of $\sigma=0.02$ m on the xy components only, with z and rotation frozen); (iii) run the diffusion agent for 12 inner SGD iterations on the candidate xy components with all 10 DDIM denoising steps differentiable and with the language conditioning removed during the optimization rollouts, scoring each candidate by the mean ℓ_2 distance of object positions at the end of the horizon from the imagined goal; and (iv) execute the best candidate for `act_horizon=10` environment steps before replanning. The environment, environment seed, 200-step horizon, and 50 rollouts per family match the plain policy protocol.

B.4 Ablation configurations

Language Table. *no AdaLN* replaces the per-DiT-block shift/scale/gate modulation with a single zero-init linear input fusion at the token-grid input, forcing conditioning to propagate through residuals only. *no per-token noise* replaces the asynchronous per-token noise schedule of Sec. 2.2 with a single shared diffusion timestep t per sample, broadcast to all valid positions; at inference, known tokens are clamped via a noised q-sample with soft blending at strength 0.5.

TraceGen. The reported model for TraceGen uses the ‘tpsqspt’ layer configuration as in Fig. 3; *no register pool* collapses the middle three layers to plain DiT at the wide dimension, removing the register-compression path and yielding the ‘tpaaapt’ layer configuration; *plain DiT* collapses all seven layers to plain DiT, removing every architectural primitive and yielding the ‘aaaaaaa’ layer configuration.

B.5 More Details

A typical WMM transformer architecture uses 8 heads with an embedding dimension of 512 and an FFN comprising a 2-layer MLP with hidden dimension 2048. The attention types are applied sequentially, one type per block. The model is trained on each application dataset using AdamW, a learning rate of $5e-5$, a weight decay factor of $1e-3$, and a decay step at 300K. For each setting, we train one set of weights on a single A6000 48GB GPU using the largest batch size that fits in memory: Language Table 64; TraceGen 4/12 for P=400/128; OMOMO 16; humanoid 8.